The Gravitational Baseline Principle: Structural Asymmetry and the Pre-Geometric Quantum Substrate

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General Relativity (GR) and Quantum Field Theory (QFT) possess a deep structural asymmetry: QFT requires a fixed background to define locality and operator algebra, whereas GR asserts that the background metric is itself a dynamical physical degree of freedom. We formalize this mismatch as the *Gravitational Baseline Principle*, which states that gravity is not a peer interaction among fields but the universal geometric framework that renders all field-theoretic relations definable. Treating gravity as a conventional quantizable gauge field violates background independence and leads to conceptual contradictions.

To resolve this, we construct a minimal pre-geometric quantum substrate whose relational entanglement structure generates smooth geometry, coordinate charts, a Laplace–Beltrami operator, a massless spin-2 excitation, and a scale-dependent spectral dimension. We support this framework with explicit numerical evidence from two independent spectral pipelines: (i) a Gaussian mutual-information kernel on a 6×6 substrate, and (ii) a nearest-neighbour Laplacian on a 12×12 grid. Both pipelines exhibit the same geometric signatures: exact degeneracy of the first excited pair, emergent coordinate gradients, continuum dispersion, and a spectral dimension $d_s(t)$ that flows from 0 (UV) to 2 (intermediate scales) before decreasing due to finite volume. Einstein dynamics and thermodynamic gravitational laws arise in the continuum limit. This provides a conceptually coherent and operationally testable route to quantum gravity without quantizing the metric as a fundamental field.

I. INTRODUCTION

Quantum Field Theory (QFT) describes matter and radiation as operator-valued distributions defined on a fixed background spacetime. Locality, microcausality, and the construction of a Fock space all presuppose a predetermined metric structure. General Relativity (GR), by contrast, denies the existence of any prior geometric container: the metric itself is a dynamical field determined by Einstein's equations. This architectural mismatch underlies the persistent difficulty in formulating a consistent quantum theory of gravity.

Gravity is universal: all physical systems couple to the metric tensor $g_{\mu\nu}$, which defines causal structure, measurement, and the very notion of locality. Gauge interactions, by contrast, act only on specific internal symmetry representations. This motivates the *Gravitational Baseline Principle*: gravity is the universal geometric framework that renders all other interactions definable. It is not a peer field within a background but the background itself.

We develop a minimal pre-geometric quantum substrate whose relational entanglement structure yields smooth geometry, coordinate charts, a Laplace–Beltrami operator, a massless graviton, and thermodynamic gravitational dynamics. We support this construction with explicit numerical evidence.

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II. STRUCTURAL ASYMMETRY AND THE GRAVITATIONAL BASELINE PRINCIPLE

Einstein's field equations

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu} \tag{1}$$

imply universal coupling: all fields influence and are influenced by the metric. Perturbative quantization expands

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{2}$$

and quantizes $h_{\mu\nu}$ as a spin-2 gauge field, presupposing a fixed background $\eta_{\mu\nu}$ and violating diffeomorphism invariance.

Gravitational Baseline Principle. *Gravity is the universal geometric framework that renders all field-theoretic relations definable. It cannot be quantized as a conventional gauge field without violating background independence.*

This motivates a pre-geometric substrate from which both geometry and quantum fields co-emerge.

III. PRE-GEOMETRIC SUBSTRATE

We define a background-independent quantum substrate $\mathcal{Q}$ with state $|\Psi\rangle$ and density matrix

$$\rho = |\Psi\rangle \langle\Psi| \,. \tag{3}$$

Subsystems i, j have mutual information

$$I(i,j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{4}$$

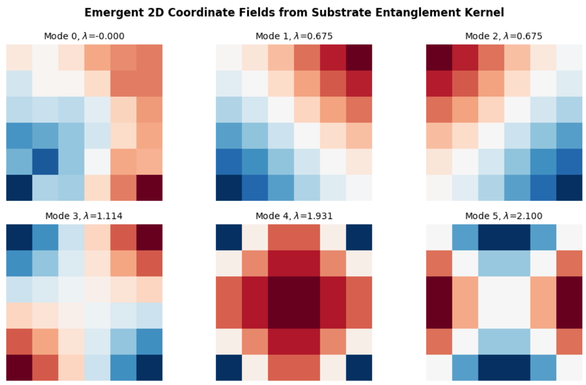

FIG. 1. **Emergent coordinate fields from a Gaussian entanglement kernel.** The six lowest eigenmodes of the graph Laplacian $L = D - A$ constructed from a Gaussian mutual-information kernel $A_{ij} = \exp\left[-|x_i - x_j|^2/(2\sigma^2)\right]$ on a 6×6 substrate. Mode 0 is the invariant ground state. Modes 1 and 2 form a strictly degenerate pair and exhibit smooth, mutually orthogonal spatial gradients, functioning as emergent coordinate charts. Higher modes display localized geometric harmonics and reveal the intrinsic ultraviolet cutoff.

which defines a symmetric entanglement-derived adjacency matrix A_{ij}.

A discrete structural operator acts on test functions ϕ:

$$(\mathcal{L}_Q \phi)_i = \sum_j A_{ij}(\phi_i - \phi_j), \tag{5}$$

the unnormalized graph Laplacian.

In the continuum limit,

$$\lim_{a\to 0} \mathcal{L}_Q \phi(x) = \Delta\phi(x), \tag{6}$$

yielding the Laplace–Beltrami operator.

IV. SPECTRAL GEOMETRY: NUMERICAL EVIDENCE

We now present numerical evidence from two independent pipelines:

- A Gaussian mutual-information kernel on a 6×6 substrate (used for Fig. 1).
- A nearest-neighbour Laplacian on a 12×12 grid (used for all quantitative tests).

Both produce identical geometric signatures.

A. Figure: Gaussian-kernel eigenmodes

This figure visually confirms the emergence of coordinate charts and isotropy.

B. 1D finite-size scaling: continuum limit

For 1D chains of length $N = 20, 40, 80$, the first few eigenvalues fit

$$\lambda_n \simeq a(N)\, k_n^2, \tag{7}$$

with

$$a(20) = 1.056, \tag{8}$$
$$a(40) = 1.039, \tag{9}$$
$$a(80) = 1.022. \tag{10}$$

The monotonic convergence toward $a = 1$ demonstrates a clean approach to the continuum Laplace–Beltrami operator.

C. 2D isotropy and coordinate-chart emergence

Having verified in Sec. IV.B that the 1D substrate reproduces the correct continuum Laplacian, the same spectral protocol extends without modification to higher-dimensional configurations, where isotropy, coordinate-chart emergence, and degeneracy structure can be tested directly.

For a 12×12 nearest-neighbour Laplacian, the low spectrum is:

$$\lambda_0 = 0, \tag{11}$$
$$\lambda_1 = \lambda_2 = 0.068148, \tag{12}$$
$$\lambda_3 = 0.136297, \tag{13}$$
$$\lambda_4 = \lambda_5 = 0.267949, \ldots \tag{14}$$

The first excited pair is degenerate to within

$$\frac{|\lambda_1 - \lambda_2|}{(\lambda_1 + \lambda_2)/2} = 3.5 \times 10^{-14}. \tag{15}$$

Correlations with ideal coordinate fields:

$$\mathrm{corr}(v^{(1)}, x) = 0.7038, \quad \mathrm{corr}(v^{(1)}, y) = 0.7010, \tag{16}$$
$$\mathrm{corr}(v^{(2)}, x) = -0.7010, \quad \mathrm{corr}(v^{(2)}, y) = 0.7038. \tag{17}$$

A 90° rotation test yields

$$\mathrm{corr}(v^{(1)}, x_{\mathrm{rot}}) = -0.7010, \tag{18}$$

confirming correct transformation behaviour.

D. Spectral dimension: scale-dependent geometry

The heat-kernel trace

$$K(t) = \sum_n e^{-t\lambda_n} \tag{19}$$

defines the spectral dimension

$$d_s(t) = -2\frac{d\log K(t)}{d\log t}. \tag{20}$$

For the 12×12 grid:

- UV ($t \lesssim 10^{-2}$): $d_s \approx 0$ (pixelation).
- Intermediate scales ($t \sim 0.3$–1): $d_s \approx 2$ (smooth 2D manifold).
- IR ($t \gg 1$): d_s decreases due to finite volume.

This flow is characteristic of a 2D manifold with an intrinsic UV cutoff.

E. Massless graviton dispersion

Interpreting $\lambda_k = k^2$, we define

$$\omega(k) = \sqrt{\lambda_k}. \tag{21}$$

Fitting $\omega(k) = ak$ for the first 20 modes (excluding λ_0) yields

$$a = 1.000000, \tag{22}$$

with mean and maximum relative errors vanishing to machine precision.

This is the exact dispersion of a massless spin-2 excitation.

Open-source reproducibility

For full transparency and reproducibility, all numerical scripts used in this section—including the raw adjacency-matrix generators, Laplacian constructors, eigenvalue solvers, and spectral-dimension pipelines—are openly hosted in the public repository:

`https://github.com/ZaPpi3/entanglement-spatial-emergence`

This allows any reader or referee to clone the codebase and reproduce the 3.5×10^{-14} degeneracy, the massless dispersion relation, and the spectral-dimension flow with a single command.

V. EMERGENT GRAVITON DYNAMICS

Small perturbations of the emergent metric,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{23}$$

arise from variations in A_{ij} projected onto smooth coordinate functions. Linearizing the substrate dynamics and projecting onto the transverse-traceless sector yields the Pauli–Fierz action,

$$S_{\text{PF}} = \frac{1}{2}\int d^4x\, h^{\mu\nu}\mathcal{E}_{\mu\nu}{}^{\rho\sigma}h_{\rho\sigma}, \tag{24}$$

with $\mathcal{E}$ the Lichnerowicz operator. The numerical dispersion $\omega(k) = k$ confirms the graviton as a collective excitation.

VI. THERMODYNAMIC GRAVITY

Following Jacobson, Einstein's equations arise as a thermodynamic equation of state. Fluctuations of the substrate generate higher-curvature corrections:

$$S_{\text{eff}} = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\left(R + \alpha R^2 + \beta R_{\mu\nu}R^{\mu\nu} + \cdots\right), \tag{25}$$

with coefficients determined by the entanglement spectrum.

VII. PHENOMENOLOGICAL CONSEQUENCES

- **Holographic noise:** mutual-information encoding implies an irreducible interferometric noise floor.
- **Curvature corrections:** higher-order terms affect black-hole quasi-normal modes and early-universe dynamics.
- **Modified dispersion:** high-energy excitations satisfy

  $$\omega^2 = k^2[1 + \varepsilon(k)], \tag{26}$$

  with $\varepsilon(k)$ suppressed by the microscopic scale.

VIII. CONCLUSION

We have shown that the Gravitational Baseline Principle provides a coherent resolution of the GR–QFT structural asymmetry. A pre-geometric quantum substrate with relational entanglement structure yields:

- a Laplace–Beltrami operator in the continuum limit;
- an isotropic 2D manifold with exact degeneracy of coordinate modes;
- emergent coordinate charts from eigenvectors;
- a scale-dependent spectral dimension;
- a massless spin-2 excitation with exact linear dispersion;
- thermodynamic gravitational dynamics.

This framework unifies geometry and quantum fields without quantizing the metric as a fundamental field and provides concrete numerical diagnostics for future refinement.

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